| Cairo University | MTHG001 | Spring 2025 |
|---|---|---|
| Faculty of Engineering | Midterm Exam | Duration: 60 minutes |

ظلل إجاباتك بوضوح في الدوائر المخصصة فقط. حافظ على نظافة ورقتك.

Shade your answers clearly in the bubble sheet only. Keep your paper clean.

1. If $A$ is an $n \times n$ nonsingular matrix, then the linear system $A\underline{X} = 2\underline{X}$:
   (a) is nonhomogeneous (b) is inconsistent
   (c) has a unique solution (d) is homogeneous (e) None

2. $A$, $B$, and $C$ are $n \times n$ nonsingular matrices. $A$ is diagonal, $B$ is lower triangular, and $C$ is upper triangular. $C(AB^T)^{-1}$ is:
   (a) diagonal (b) upper triangular
   (c) symmetric (d) lower triangular (e) None

3. If $A$ is an $n \times n$ symmetric matrix, then $(A + A^2)$ is:
   (a) skew-symmetric (b) diagonal
   (c) symmetric (d) triangular (e) None

4. If $A = \begin{bmatrix} 2 & 1 & -6 \\ -1 & 3 & r \end{bmatrix}$, $r \in \mathbb{R}$, then the entry in row 2 and column 3 of $A^TA$ is equal to:
   (a) $-6 + 3r$ (b) $6 - 3r$ (c) $-12 - r$ (d) $-12 + r$ (e) None

5. The linear system $\begin{bmatrix} 1 & 2 & 5 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \\ 0 & 1 & 4 \end{bmatrix} \underline{X} = \begin{bmatrix} 4 \\ 0 \\ 3 \\ w \end{bmatrix}$ will be inconsistent if $w$ is not equal to:
   (a) 9 (b) 6 (c) 3 (d) $-3$ (e) None

6. If $(A^{-1} + 2I)^T = \begin{bmatrix} 1 & 7 \\ 0 & g \end{bmatrix}$, $g \neq 2$, then the entry in row 2 and column 1 of $A$ is equal to:
   (a) $-7/(g-2)$ (b) $1/(g-2)$
   (c) $-1/(g-2)$ (d) $7/(g-2)$ (e) None

7. Consider the linear system $\begin{aligned} x + y - z &= a \\ 2x + 5y + 2z &= 2 \\ 3x + 2y + z &= 8 \end{aligned}$. If $x = r$, $y = s$, and $z = 3$ is the solution of this system, where $r, s, a \in \mathbb{R}$, then $a$ is equal to:
   (a) $-2$ (b) 2 (c) 3 (d) $-3$ (e) None

8. Consider the linear system $\begin{aligned} x + 4y + 3z &= 2 \\ x + 5y + z &= 5 \\ 3x + 2y + 29z &= -24 \end{aligned}$. If $z = t, t \in \mathbb{R}$, then $x$ is equal to:
   (a) $-10 + 11t$ (b) $-5 + 4t$ (c) $5 - 4t$ (d) $-10 - 11t$ (e) None

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9. The linear system $\begin{aligned} x - 4y + 2z &= 5 \\ x + 3y + 7z &= 6 \\ rx + sy + tz &= 7 \end{aligned}$ will not have a unique solution if:

(a) $r - 2s + 3t = 0$ (b) $30r + 4s - 7t = 0$

(c) $34r + 5s - 7t = 0$ (d) $2r + 5s + 2t = 0$ (e) None

10. If the matrix $\begin{bmatrix} 1 & 3 & 4 - f^2 & 2 & 0 & 4 \\ 0 & 0 & 1 & 3 & f - 2 & 1 \\ 0 & 0 & 0 & 0 & 1 & 2 \end{bmatrix}$ is in reduced row echelon form (RREF), then $f$ is equal to:

(a) $-2$ (b) 2 (c) 3 (d) $-1$ (e) None

Consider $E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{bmatrix}$.

11. The elementary row operation performed on a $3 \times 6$ matrix when it is multiplied from the left by $E$ is:

(a) $3r_3 + r_2 \to r_2$ (b) $-3r_2 + r_3 \to r_3$

(c) $-3r_3 + r_2 \to r_2$ (d) $3r_2 + r_3 \to r_3$ (e) None

12. The entry in row 2 and column 3 of $E^{-1}$ is:

(a) 3 (b) $-3$ (c) 1/3 (d) $-1/3$ (e) None

13. If $A, B, C$, and $D$ are $n \times n$ nonsingular matrices, and $(XA)^T + B = C$, then $X$ is equal to:

(a) $A^{-1}(C - B)^T$ (b) $(C + B)^T A^{-1}$

(c) $A^{-1}(C + B)^T$ (d) $(C - B)^T A^{-1}$ (e) None

14. Given two matrices $A$ and $B$, such that $AB$ and $BA$ are defined, then $AB$ is a square matrix.

(a) False (b) True

15. Consider the LU decomposition of the matrix $\begin{bmatrix} 2 & 2 & 4 \\ 3 & 2 & 3 \\ 4 & 5 & f \end{bmatrix}$, where $|f| > 12$.

The entry in row 3 and column 3 of $L$ is equal to:

(a) $f - 12$ (b) $1/(f - 11)$ (c) $f - 11$ (d) $1/(f - 12)$ (e) None

16. The linear system $\begin{aligned} x - 2y + 4z &= a \\ 2x - 3y + 11z &= b \\ 3x - 13y - 9z &= c \end{aligned}$ will be consistent if:

(a) $c = 17a - 7b$ (b) $c = 15a - 6b$

(c) $c = -17a + 7b$ (d) $c = -15a + 6b$ (e) None

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Given the linear system
$$\begin{aligned} x + y + z &= 6 \\ x + 2y + z &= 4; \\ x + y + kz &= b \end{aligned}$$

17. The system has infinitely many solutions when:
    (a) $k = 1$ and $b = 6$ (b) $k = 4$ and $b = 2$
    (c) $k = 1$ and $b = 2$ (d) $k = 2$ and $b = 4$ (e) None

18. The system has no solution when:
    (a) $k = 1$ and $b = 6$ (b) $k = 4$ and $b = 2$
    (c) $k = 1$ and $b = 2$ (d) $k = 2$ and $b = 4$ (e) None

Consider the LU decomposition of the matrix $\begin{bmatrix} 2 & 1 & 2 & 0 \\ 2 & 2 & 3 & 2 \\ 4 & 3 & 6 & 3 \\ 0 & 2 & 3 & -6 \end{bmatrix}$.

19. The entry in row 4 and column 4 of $L$ is equal to:
    (a) $-1/11$ (b) $-11$ (c) $-10$ (d) $-1/10$ (e) None

20. The entry in row 3 and column 4 of $U$ is equal to:
    (a) $-1$ (b) $-3$ (c) 3 (d) 1 (e) None

Consider $A = \begin{bmatrix} 1 & 3 & 2 & 5 & 1 & 6 \\ 3 & 10 & 5 & 17 & 6 & 22 \\ 4 & 14 & 6 & 25 & 9 & 38 \\ 2 & 5 & 5 & 6 & 2 & 5 \end{bmatrix}$. Let $R$ be the reduced row echelon form (RREF) of $A$.

21. The entry in row 2 and column 6 of $R$ is equal to:
    (a) $-50$ (b) $-47$ (c) $-53$ (d) $-44$ (e) None

22. The entry in row 1 and column 6 of $R$ is equal to:
    (a) 81 (b) 73 (c) 65 (d) 57 (e) None

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# Midterm—Algebra
# Spring 2025

Algebra Team (2025)

October 30, 2025

**1. If $A$ is an $n \times n$ nonsingular matrix, then the linear system $A\mathbf{x} = 2\mathbf{x}$:**

**Answer: (d) is homogeneous**; the wrong answer would be to assume that $\mathbf{x} = 2A^{-1}\mathbf{x}$ and conclude that it has a unique solution.

To do it correctly, rearrange the equation

$$A\mathbf{x} - 2\mathbf{x} = \underbrace{(A - 2I)}_{\substack{\text{coefficient} \\ \text{matrix}}} \mathbf{x} = \mathbf{0}$$

and we've successfully concluded that the system $A\mathbf{x} = 2\mathbf{x}$ must be homogeneous.

**2. If $A$, $B$ and $C$ are $n \times n$ nonsingular matrices. $A$ is diagonal, $B$ is lower triangular and $C$ is upper triangular, then $C(AB^T)^{-1}$:**

**Answer: (b) upper triangular**; firstly, since $B$ is lower triangular, then $B^T$ is upper triangular. Secondly, since $A$ is a diagonal matrix, then $AB^T$ is upper triangular. To understand that better, it could be helpful to look at how they multiply:

$$\begin{bmatrix} a_1 & 0 & \dots & 0 \\ 0 & a_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & a_n \end{bmatrix} \begin{bmatrix} b_{11} & b_{21} & \dots & b_{n1} \\ 0 & b_{22} & \dots & b_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & b_{nn} \end{bmatrix} = \begin{bmatrix} a_1b_{11} & a_1b_{21} & \dots & a_!b_{n1} \\ 0 & a_2b_{22} & \dots & a_2b_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & a_nb_{nn} \end{bmatrix}$$

and third, the inverse of an upper triangular matrix is still upper triangular, so $(AB^T)^{-1}$ must be upper triangular. And then the multiplication of upper triangular matrices gives another upper triangular matrix. Finally, $C(AB^T)^{-1}$ must be upper triangular.

**3. If $A$ is an $n \times n$ symmetric matrix, then $(A^2 + A)$ is:**

**Answer: (c) symmetric**; let $D$ be a matrix, then

(a) if $D$ is skew-symmetric; $D^T = -D$

(b) if $D$ is symmetric; $D^T = D$

Then take the transpose of $A^2 + A$

$$(A^2 + A)^T = (A^2)^T + A^T = A^2 + A \qquad (A^T = A \text{ and } (A^n)^T = (A^T)^n)$$

so, $A^2 + A$ is symmetric.

4. If $A = \begin{bmatrix} 2 & 1 & -6 \\ -1 & 3 & r \end{bmatrix}$, $r \in \mathbb{R}$, then the entry in row 2 and column 3 of $A^T A$ is:

**Answer: (a)** $-6 + 3r$; firstly, $A^T$

$$A^T = \begin{bmatrix} 2 & -1 \\ 1 & 3 \\ -6 & r \end{bmatrix}$$

then, $A^T A$ can be carried out readily

$$\begin{bmatrix} 2 & -1 \\ 1 & 3 \\ -6 & r \end{bmatrix} \begin{bmatrix} 2 & 1 & -6 \\ -1 & 3 & r \end{bmatrix}$$

to save time, we notice they require the entry in **row 2** and **column 3**, then we multiply row 2 of the first matrix by column 3 of the second matrix

$$1 \times -6 + 3 \times r = -6 + 3r$$

5. The linear system $\begin{bmatrix} 1 & 2 & 5 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \\ 0 & 1 & 4 \end{bmatrix} \mathbf{x} = \begin{bmatrix} 4 \\ 0 \\ 3 \\ w \end{bmatrix}$ will be inconsistent if $w$ is not equal to:

**Answer: (b)** 6; we proceed using the augmented matrix, performing Gaussian elimination on it

$$\left[\begin{array}{ccc|c} 1 & 2 & 5 & 4 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 1 & 3 \\ 0 & 1 & 4 & w \end{array}\right]$$

$$\xrightarrow{r_4 - r_2 \to r_4} \left[\begin{array}{ccc|c} 1 & 2 & 5 & 4 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 2 & w \end{array}\right]$$

$$\xrightarrow{r_4 - 2r_3 \to r_4} \left[\begin{array}{ccc|c} 1 & 2 & 5 & 4 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & w - 6 \end{array}\right]$$

then, for the system to be inconsistent:

$$\#\text{leading variables} = 3 < \#\text{non-zero rows}$$

so $w - 6 \neq 0 \Rightarrow w \neq 6$.

6. If $(A^{-1}+2I)^T = \begin{bmatrix} 1 & 7 \\ 0 & g \end{bmatrix}$, $g \neq 2$, then the entry in row 2 and column 1 of $A$ is:

**Answer: (d)** $7/(g-2)$; firstly take the transpose and then add $-2I$ to both sides

$$A^{-1} = \begin{bmatrix} 1 & 0 \\ 7 & g \end{bmatrix} - 2\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 7 & g-2 \end{bmatrix}$$

then take the inverse of both sides

$$A = \frac{1}{2-g}\begin{bmatrix} g-2 & 0 \\ -7 & -1 \end{bmatrix}$$

then the entry in row 2 and column 1 is $-7/(2-g) = 7/(g-2)$.

7. Consider the linear system $\begin{cases} x+y-z=a \\ 2x+5y+2z=2 \\ 3x+2y+z=8 \end{cases}$. If $x=r$, $y=s$ and $z=3$ is the solution to this system, where $r$, $s$, $a \in \mathbb{R}$, then $a$ is equal to:

**Answer: (a)** $-2$; putting the system in augmented matrix form

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & a \\ 2 & 5 & 2 & 2 \\ 3 & 2 & 1 & 8 \end{array}\right]$$

then proceeding using Gaussian elimination

$$\xrightarrow[r_3-3r_1 \to r_3]{r_2-2r_1 \to r_2} \left[\begin{array}{ccc|c} 1 & 1 & -1 & a \\ 0 & 3 & 4 & 2-2a \\ 0 & -1 & 4 & 8-3a \end{array}\right]$$

$$\xrightarrow{(1/3)r_2 \to r_2} \left[\begin{array}{ccc|c} 1 & 1 & -1 & a \\ 0 & 1 & 4/3 & (2-2a)/3 \\ 0 & -1 & 4 & 8-3a \end{array}\right]$$

$$\xrightarrow{r_3+r_2 \to r_3} \left[\begin{array}{ccc|c} 1 & 1 & -1 & a \\ 0 & 1 & 4/3 & (2-2a)/3 \\ 0 & 0 & 16/3 & (26-11a)/3 \end{array}\right]$$

then using the last row, $16z/3 = (26-11a)/3 \Rightarrow a = (26-16z)/11$, substitute $z=3$, then $a=-2$.

8. Consider the linear system $\begin{cases} x+4y+3z=2 \\ x+5y+z=5 \\ 3x+2y+29z=-24 \end{cases}$. If $z=t$, $t \in \mathbb{R}$, then $x$ is equal to:

**Answer: (d)** $-10-11t$; putting the system in augmented matrix form

$$\left[\begin{array}{ccc|c} 1 & 4 & 3 & 2 \\ 1 & 5 & 1 & 5 \\ 3 & 2 & 29 & -24 \end{array}\right]$$

then proceeding using Gaussian elimination

$$\xrightarrow[r_3-3r_1 \to r_3]{r_2-r_1 \to r_2} \left[\begin{array}{ccc|c} 1 & 4 & 3 & 2 \\ 0 & 1 & -2 & 3 \\ 0 & -10 & 20 & -30 \end{array}\right]$$

$$\xrightarrow{r_3+10r_2 \to r_3} \left[\begin{array}{ccc|c} 1 & 4 & 3 & 2 \\ 0 & 1 & -2 & 3 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

then, using the second row $y - 2t = 3 \Rightarrow y = 3 + 2t$. And the first row $x + 4(3 + 2t) + 3t = 2 \Rightarrow x = -10 - 11t$.

9. The linear system $\begin{cases} x - 4y + 2z = 5 \\ x + 3y + 7z = 6 \\ rx + sy + tz = 7 \end{cases}$ will not have a unique solution if:

**Answer: (c)** $34r + 5s - 7t = 0$; if we put our system into matrix equation form

$$\begin{bmatrix} 1 & -4 & 2 \\ 1 & 3 & 7 \\ r & s & t \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 5 \\ 6 \\ 7 \end{bmatrix}$$

the system will have one of three possible solutions:

i. a unique solution (the determinant is nonzero!)

ii. infinitely many solutions.

iii. no solution.

then for our system **to not have** a unique solution, the coefficient matrix's determinant has to equal zero. (calculating the determinant using the final row)

$$\begin{vmatrix} 1 & -4 & 2 \\ 1 & 3 & 7 \\ r & s & t \end{vmatrix} = r \times (-28 - 6) - s \times (7 - 2) + t \times (3 + 4) = -34r - 5s + 7t = 0$$

then $34r + 5s - 7t = 0$.

10. If the matrix $\begin{bmatrix} 1 & 3 & 4 - f^2 & 2 & 0 & 4 \\ 0 & 0 & 1 & 3 & f - 2 & 1 \\ 0 & 0 & 0 & 0 & 1 & 2 \end{bmatrix}$ is in RREF, then $f$ is equal to:

**Answer: (b)** 2; RREF satisfies four conditions:

i. any zero-rows $\begin{bmatrix} 0 & 0 & \dots & 0 \end{bmatrix}$ appear at the very bottom of the matrix.

ii. the first non-zero entry from the left in any row is a 1 (called a leading one.)

iii. In any two successive rows that do not consist entirely of zeros, the leading 1 in the lower row occurs farther to the right than the leading 1 in the higher row (staircase pattern.)

iv. In a column that contains a leading 1, all other entries are zeros (only non-zero entry in the column.)

then, if we look for our leading ones

$$\begin{bmatrix} \boxed{1} & 3 & 4-f^2 & 2 & 0 & 4 \\ 0 & 0 & \boxed{1} & 3 & f-2 & 1 \\ 0 & 0 & 0 & 0 & \boxed{1} & 2 \end{bmatrix}$$

then to satisfy property (iv), we need to zero out above our second and third leading one. Then $f-2=0 \Rightarrow f=2$, and that means $4-f^2=4-4=0$.

Consider $E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{bmatrix}$

11. The elementary row operation performed on a $3 \times 6$ matrix when it is multiplied from the left by $E$ is:

**Answer: (c)** $-3r_3 + r_2 \to r_2$; the elementary row operation is that which is done on the original matrix. To get the elementary matrix from the operation, we apply the same operation on the identity matrix $I$.

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{r_2 - 3r_3 \to r_2} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{bmatrix}$$

Additionally, if the number lies on the main diagonal, then the operation is multiplying that number by the row it is in.

If it is off the main diagonal, look which 1 is in the same column as it. Then, add that number times the row of the 1 in the same column to the row the number is in like here. $-3$ lies in the same column as the 1 in row 3, so the operation is $-3r_3 + r_2 \to r_2$.

12. The entry in row 2 and column 3 of $E^{-1}$ is:

**Answer: (a)** 3; the inverse elementary matrix is that which reverses the effect of the original elementary matrix.

In our case, $E$ added -3 times the third row to the second row. To reverse that, add 3 times the third row to the second row, which means

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}$$

so that the entry in row 2 and column 3 is 3.

13. If $A$, $B$ and $C$ are $n \times n$ nonsingular matrices, and $(XA)^T + B = C$, then $X$ is equal to:

**Answer: (d)** $(C-B)^T A^{-1}$;

$$
\begin{aligned}
(XA)^T &= C - B \\
XA &= (C-B)^T \\
X &= (C-B)^T A^{-1}
\end{aligned}
$$

14. Given two matrices $A$ and $B$, such that $AB$ and $BA$ are defined, then $AB$ is a square matrix:

**Answer: (b) True**; if we let $A$ and $B$ be of sizes $n \times m$ and $m \times r$ respectively, then

(a) $A_{n\times m}B_{m\times r}$ is defined.

(b) $B_{m\times r}A_{n\times m}$ is not defined.

obviously then, for $BA$ to be defined $r$ has to equal $n$. Then (a) produces a square matrix of size $n \times n$ which is what we wanted to prove.

15. Consider the LU decomposition of the matrix $\begin{bmatrix} 2 & 2 & 4 \\ 3 & 2 & 3 \\ 4 & 5 & f \end{bmatrix}$, where $|f| > 12$. The entry in row 3 and column 3 of $L$ is equal to:

**Answer: (c)** $f - 11$; using the careful book-keeping method

$$
\xrightarrow{(1/2)r_1 \to r_1} \begin{bmatrix} 1 & 1 & 2 \\ 3 & 2 & 3 \\ 4 & 5 & f \end{bmatrix} \qquad L = \begin{bmatrix} 2 & 0 & 0 \\ & & 0 \\ & & \end{bmatrix}
$$

$$
\xrightarrow[r_3 - 4r_1 \to r_3]{r_2 - 3r_1 \to r_2} \begin{bmatrix} 1 & 1 & 2 \\ 0 & -1 & -3 \\ 0 & 1 & f-8 \end{bmatrix} \qquad L = \begin{bmatrix} 2 & 0 & 0 \\ 3 & & 0 \\ 4 & & \end{bmatrix}
$$

$$
\xrightarrow{-r_2 \to r_2} \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 3 \\ 0 & 1 & f-8 \end{bmatrix} \qquad L = \begin{bmatrix} 2 & 0 & 0 \\ 3 & -1 & 0 \\ 4 & & \end{bmatrix}
$$

$$
\xrightarrow{r_3 - r_2 \to r_3} \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & f-11 \end{bmatrix} \qquad L = \begin{bmatrix} 2 & 0 & 0 \\ 3 & -1 & 0 \\ 4 & 1 & \end{bmatrix}
$$

$$
\xrightarrow{1/(f-11)r_3 \to r_3} \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix} \qquad L = \begin{bmatrix} 2 & 0 & 0 \\ 3 & -1 & 0 \\ 4 & 1 & f-11 \end{bmatrix}
$$

then the entry in row 3 and column 3 of $L$ is equal to $f - 11$.

16. The linear system $\begin{cases} x - 2y + 4z = a \\ 2x - 3y + 11z = b \\ 3x - 13y - 9z = c \end{cases}$ will be consistent if:

**Answer: (a)** $c = 17a - 7b$; putting our system into augmented matrix form

$$\left[\begin{array}{ccc|c} 1 & -2 & 4 & a \\ 2 & -3 & 11 & b \\ 3 & -13 & -9 & c \end{array}\right]$$

then proceeding using Gaussian elimination

$$\xrightarrow[r_3 - 3r_1 \to r_3]{r_2 - 2r_1 \to r_2} \left[\begin{array}{ccc|c} 1 & -2 & 4 & a \\ 0 & 1 & 3 & b - 2a \\ 0 & -7 & -21 & c - 3a \end{array}\right]$$

$$\xrightarrow{r_3 + 7r_2 \to r_3} \left[\begin{array}{ccc|c} 1 & -2 & 4 & a \\ 0 & 1 & 3 & b - 2a \\ 0 & 0 & 0 & c - 17a + 7b \end{array}\right]$$

it is apparent that the system can only have infinitely many solutions (consistent) if the final row has zeroes everywhere; $c - 17a + 7b = 0 \Rightarrow c = 17a - 7b$.

Given the linear system $\begin{cases} x + y + z = 6 \\ x + 2y + z = 4 \\ x + y + kz = b \end{cases}$ :

17. The system has infinitely many solutions when:

**Answer: (a)** $k = 1$, $b = 6$; putting our system into augmented matrix form

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & k & b \end{array}\right]$$

then proceeding with Gaussian elimination

$$\xrightarrow[r_3 - r_1 \to r_3]{r_2 - r_1 \to r_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & k - 1 & b - 6 \end{array}\right]$$

then, if the system has infinitely many solutions, the final row must be all zeroes; $k - 1 = 0 \Rightarrow k = 1$ and $b - 6 = 0 \Rightarrow b = 6$.

18. The system has no solution when:

**Answer: (c)** $k = 1$, $b = 2$; from the previous problem, the system has no solution if and only if the final row is all zeroes except the augmented column. $k - 1 = 0 \Rightarrow k = 1$ and $b - 6 \neq 0 \Rightarrow b \neq 6$.

Consider the LU decomposition of the matrix $\begin{bmatrix} 2 & 1 & 2 & 0 \\ 2 & 2 & 3 & 2 \\ 4 & 3 & 6 & 3 \\ 0 & 2 & 3 & -6 \end{bmatrix}$:

19. The entry in row 4 and column 4 of $L$ is equal to:

**Answer: (b)** $-11$; using the careful book-keeping method

$$\xrightarrow{(1/2)r_1 \to r_1} \begin{bmatrix} 1 & 1/2 & 1 & 0 \\ 2 & 2 & 3 & 2 \\ 4 & 3 & 6 & 3 \\ 0 & 2 & 3 & -6 \end{bmatrix} \qquad L = \begin{bmatrix} 2 & 0 & 0 & 0 \\ & & 0 & 0 \\ & & & 0 \\ & & & \end{bmatrix}$$

$$\xrightarrow[r_3 - 4r_1 \to r_3]{r_2 - 2r_1 \to r_2} \begin{bmatrix} 1 & 1/2 & 1 & 0 \\ 0 & 1 & 1 & 2 \\ 0 & 1 & 2 & 3 \\ 0 & 2 & 3 & -6 \end{bmatrix} \qquad L = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 2 & & 0 & 0 \\ 4 & & & 0 \\ 0 & & & \end{bmatrix}$$

$$\xrightarrow[r_4 - 2r_2 \to r_4]{r_3 - r_2 \to r_3} \begin{bmatrix} 1 & 1/2 & 1 & 0 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & -10 \end{bmatrix} \qquad L = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ 4 & 1 & & 0 \\ 0 & 2 & & \end{bmatrix}$$

$$\xrightarrow{r_4 - r_3 \to r_4} \begin{bmatrix} 1 & 1/2 & 1 & 0 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & -11 \end{bmatrix} \qquad L = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ 4 & 1 & 1 & 0 \\ 0 & 2 & 1 & \end{bmatrix}$$

$$\xrightarrow{(-1/11)r_4 \to r_4} \begin{bmatrix} 1 & 1/2 & 1 & 0 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \qquad L = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ 4 & 1 & 1 & 0 \\ 0 & 2 & 1 & -11 \end{bmatrix}$$

then the entry in row 4 and column 4 of $L$ is $-11$.

20. The entry in row 3 and column 4 of $U$ is equal to:

**Answer: (d)** 1; from the previous problem the entry in row 3 and column 4 is 1, where $U$ is

$$\begin{bmatrix} 1 & 1/2 & 1 & 0 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Consider $A = \begin{bmatrix} 1 & 3 & 2 & 5 & 1 & 6 \\ 3 & 10 & 5 & 17 & 6 & 22 \\ 4 & 14 & 6 & 25 & 9 & 38 \\ 2 & 5 & 5 & 6 & 2 & 5 \end{bmatrix}$. Let $R$ be the reduced row echelon form of $A$.

21. The entry in row 2 and column 6 of $R$ is equal to:

**Answer:** **(c)** $-53$; performing row reduction on our matrix

$$\xrightarrow[r_4-2r_1 \to r_4]{r_2-3r_1 \to r_2,\ r_3-4r_1 \to r_3} \begin{bmatrix} 1 & 3 & 2 & 5 & 1 & 6 \\ 0 & 1 & -1 & 2 & 3 & 4 \\ 0 & 2 & -2 & 5 & 5 & 14 \\ 0 & -1 & 1 & -4 & 0 & -7 \end{bmatrix}$$

$$\xrightarrow[r_4+r_2 \to r_4]{r_1-3r_2 \to r_1,\ r_3-2r_2 \to r_3} \begin{bmatrix} 1 & 0 & 5 & -1 & -8 & -6 \\ 0 & 1 & -1 & 2 & 3 & 4 \\ 0 & 0 & 0 & 1 & -1 & 6 \\ 0 & 0 & 0 & -2 & 3 & -3 \end{bmatrix}$$

$$\xrightarrow[r_4+2r_3 \to r_4]{r_1+r_3 \to r_1,\ r_2-2r_3 \to r_2} \begin{bmatrix} 1 & 0 & 5 & 0 & -9 & 0 \\ 0 & 1 & -1 & 0 & 5 & -8 \\ 0 & 0 & 0 & 1 & -1 & 6 \\ 0 & 0 & 0 & 0 & 1 & 9 \end{bmatrix}$$

$$\xrightarrow[r_3+r_4 \to r_3]{r_1+9r_4 \to r_1,\ r_2-5r_4 \to r_2} \begin{bmatrix} 1 & 0 & 5 & 0 & 0 & 81 \\ 0 & 1 & -1 & 0 & 0 & -53 \\ 0 & 0 & 0 & 1 & 0 & 15 \\ 0 & 0 & 0 & 0 & 1 & 9 \end{bmatrix}$$

then the entry in row 2 and column 6 is $-53$.

22. The entry in row 1 and column 6 of $R$ is equal to:

**Answer:** **(a)** 81; from the previous problem, the entry in row 1 and column 6 of $R$ is 81.